Beyond Surface Affect: Why Sentiment Detection Alone is Insufficient for Intent Interpretation in Human–AI Communication

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Abstract

Sentiment detection—the automated classification of affective polarity in natural language text—has achieved strong empirical performance across standard benchmarks and is widely deployed in conversational AI systems. However, in interactive Human–AI communication contexts such as interview coaching assistants, reflective writing platforms, and mental health dialogue agents, sentiment polarity alone is demonstrably insufficient for interpreting user intent. This paper formalizes the distinction between sentiment detection and intent interpretation using mathematical notation. We define a sentiment classifier $S(x)$ operating over surface text and contrast it with a contextualized intent function $I(x, C, G, H)$, where C denotes dialogue context, G encodes user goals, and H represents interaction history. We prove formally that $S(x)$ does not imply $I(x, C, G, H)$ in the general case, and identify four canonical failure modes of sentiment-only systems. We present system-level case observations from MindNook, a sentiment-aware reflective diary platform, to ground these theoretical distinctions empirically. We propose a Multi-Layer Intent Modeling (MLIM) framework integrating affective, pragmatic, goal-theoretic, and contextual components, and conclude with an open research agenda addressing context management, multi-modal grounding, and privacy-preserving personalization.

Keywords: intent modeling; sentiment analysis; natural language understanding; affective computing; dialogue systems; Human–AI interaction; pragmatics; context-aware NLP; conversational AI; speech act theory

1 Introduction

The proliferation of conversational AI systems across professional, educational, and mental health domains has created growing demand for models capable of accurately interpreting what users *mean*, not merely what they *feel*. Sentiment analysis—the task of extracting affective orientation (positive, negative, or neutral) from text—has long served as a foundation for understanding user state in such systems [24, 30]. State-of-the-art transformer-based models achieve high accuracy on established affective benchmarks [39, 8], reinforcing a widespread assumption that affect-aware systems are also intent-aware.

This assumption is structurally incorrect. Sentiment and intent are formally distinct constructs. Consider a user who writes "That was surprisingly easy" during an AI-mediated interview preparation session. A sentiment classifier will register positive polarity. An intent-aware model, however, must recognize that the utterance may simultaneously encode concern about underperformance, a request for increased challenge, or a masked anxiety about competency. The surface-level affect—positive—provides no discriminative information for resolving among these interpretations. This divergence is not an edge case; it is a structural property of natural human communication rooted in Gricean pragmatics [16] and speech act theory [1, 38].

The practical consequences of conflating sentiment with intent are significant. A system that responds to positive sentiment with affirmation when the user's intent encodes a request for deeper engagement fails communicatively. A system that responds to negative sentiment with comfort when the intent is goal-directed critique fails instrumentally. These mismatches erode user trust, reduce system utility, and—in high-stakes contexts such as mental health support—may cause direct harm.

This paper makes the following contributions:

1. We provide formal mathematical definitions of sentiment detection and intent interpretation, establishing a precise theoretical vocabulary.
2. We prove that $S(x)$ does not imply $I(x, C, G, H)$ in the general case, with a formal proposition and proof sketch.
3. We enumerate four canonical failure modes arising when intent modeling is reduced to sentiment scoring.
4. We present grounded system-level case observations from MindNook.
5. We propose the Multi-Layer Intent Modeling (MLIM) framework.
6. We outline a structured open research agenda.

We emphasize at the outset that sentiment detection is not superfluous. Affective signals are a *necessary* component of any intent-aware system. Our argument is strictly that they are *insufficient*—full intent interpretation requires additional formal machinery and contextual grounding beyond polarity classification.

2 Background and Related Work

2.1 Sentiment Analysis: Methods and Limits

Sentiment analysis has a well-established research trajectory. Early lexicon-based approaches—including SentiWordNet [2] and VADER [20]—mapped words to affective scores, enabling interpretable but context-insensitive polarity detection. Machine learning approaches applied support vector machines and Naive Bayes classifiers to review corpora [31]. Deep learning methods using CNNs [21] and LSTMs [41] improved accuracy on sentence-level sentiment tasks. Pre-trained transformer models—BERT [8] and RoBERTa [26]—subsequently achieved near-human performance on benchmarks such as SST-2 [39] and SemEval affective tasks [33].

Aspect-based sentiment analysis (ABSA) extended the paradigm by linking polarity to specific opinion targets [49, 40], achieving fine-grained affect extraction. Despite these advances, the fundamental operation remains polarity attribution: given utterance x, produce a label or scalar indicating affective orientation. Context beyond the sentence boundary is typically limited and goal-agnostic.

2.2 Intent Detection and Dialogue Understanding

Intent detection in task-oriented dialogue has been studied since the ATIS corpus [35] and SLU benchmarks [42]. Neural classifiers using CNNs and BiLSTMs were proposed by Liu and Lane [25], while joint modeling of intent and slot filling using BERT-based encoders was addressed by Zhang et al. [51] and Xu and Hu [48]. The CLINC150 benchmark [23] introduced out-of-scope intent detection, highlighting open-world intent challenges. Efficient intent detection using sentence transformers was explored by Casanueva et al. [7]. Large language models [6, 45] have demonstrated in-context intent recognition without task-specific fine-tuning, though their behavior under long interaction histories and complex user goals remains incompletely understood.

Critically, the predominant framing of intent detection in the literature has been as a classification problem over a closed label set—adequate for narrow task-oriented systems but insufficient for open-domain Human–AI communication where user goals are latent, evolving, and often unspoken.

2.3 Pragmatics and Speech Act Theory

The theoretical foundation for distinguishing surface meaning from communicative intent originates in the philosophy of language. Austin [1] proposed the distinction between locutionary acts (literal propositional content), illocutionary acts (communicative force, e.g., requesting, asserting, promising), and perlocutionary acts (effect produced in the listener). Searle [38] systematized this into a taxonomy of speech acts: directives, commissives, expressives, declaratives, and representatives. Grice [16] introduced the Cooperative Principle and conversational maxims of quantity, quality, relation, and manner, establishing that speakers routinely convey more than their words literally state through *implicature*. Computational pragmatics [4] has sought

to operationalize these insights for NLP, though systematic integration into deployed dialogue systems remains limited.

2.4 Affective Computing and Human–AI Interaction

Affective computing [32] established the program of equipping machines to recognize and interpret human affect. Applications include emotionally adaptive agents [5], affective tutoring systems [9], and mental health chatbots [13, 43]. The critical limitation documented across this literature is the use of affective state as a proxy for user need—adequate in narrow domains but unreliable in complex, longitudinal interactions where goals evolve and affect masks intent. Baltrusaitis et al. [3] survey multimodal machine learning for affective computing, emphasizing fusion strategies across linguistic, acoustic, and visual modalities. Poria et al. [34] review conversational emotion recognition, noting that context substantially alters emotional interpretation.

3 Problem Formalization

3.1 Notation and Definitions

Let $\mathcal{X}$ denote the space of natural language utterances over vocabulary V. Let $\mathcal{A} = \{\text{positive}, \text{negative}, \text{neutral}\}$ denote the affective label space, and let $\mathcal{I}$ denote the intent representation space (defined in Section 3.3).

Definition 3.1 (Utterance)**.** An utterance $x \in \mathcal{X}$ is a finite sequence of tokens $x = (w_1, w_2, \ldots, w_n)$ where each $w_i \in V$.

Definition 3.2 (Dialogue Context)**.** Let $C = (x_{t-k}, \ldots, x_{t-1})$ denote the ordered sequence of the k most recent preceding utterances in a dialogue session, constituting the local context window, where $k \in \mathbb{N}$ is a configurable context horizon.

Definition 3.3 (User Goal)**.** Let $G \in \mathcal{G}$ denote the user's communicative goal, where $\mathcal{G}$ is a structured representation of the user's task objective, interaction purpose, and success criteria. G may be partially observable or latent.

Definition 3.4 (Interaction History)**.** Let $H = \{(x_\tau, r_\tau, a_\tau) : \tau = 1, \ldots, t-1\}$ denote the full interaction history up to time step t, where x_τ is the user utterance at step τ, r_τ is the system response, and a_τ is the action taken.

3.2 Formal Definition of Sentiment Detection

Definition 3.5 (Sentiment Classifier)**.** A sentiment classifier is a function $S : \mathcal{X} \to \mathcal{A}$, or in the probabilistic formulation:

$$S : \mathcal{X} \to \Delta(\mathcal{A}) \tag{1}$$

where $\Delta(\mathcal{A})$ denotes the probability simplex over $\mathcal{A}$. The predicted sentiment is:

$$\hat{s} = \arg\max_{a \in \mathcal{A}} P(a \mid x; \theta_S) \tag{2}$$

where θ_S are the learned parameters of the sentiment model. In the continuous-valued variant [37], S maps to a point in $\mathbb{R}^d$ where $d = 2$ for the valence–arousal space.

Observation 3.1. In all formulations, $S(x)$ is a function of x *alone*. It does not condition on C, G, or H.

3.3 Formal Definition of Intent Interpretation

Definition 3.6 (Intent Function)**.** An intent interpretation function is defined as:

$$I : \mathcal{X} \times \mathcal{C} \times \mathcal{G} \times \mathcal{H} \;\rightarrow\; \Delta(\mathcal{I}) \tag{3}$$

The predicted intent at time t is:

$$\hat{\imath}_t = \arg\max_{i \in \mathcal{I}} \; P(i \mid x_t,\, C_t,\, G,\, H_t;\, \theta_I) \tag{4}$$

where θ_I are the parameters of the intent model.

The intent space $\mathcal{I}$ may take several forms: (i) a closed label set (e.g., `BookFlight`, `RequestInfo`) for task-oriented dialogue; (ii) a structured tuple (act_type, topic, urgency, confidence) for open-domain systems; or (iii) a latent embedding space for parametric intent representation.

Definition 3.7 (Intent-Relevant Context)**.** The intent-relevant context at time t is the tuple:

$$\Omega_t = (x_t,\, C_t,\, G,\, H_t) \tag{5}$$

Intent interpretation is thus a mapping $I : \Omega_t \rightarrow \Delta(\mathcal{I})$.

3.4 Formal Problem Statement

Problem Statement. Given a deployed Human–AI system that implements $S(x)$ as its primary user-understanding mechanism, determine whether $S(x)$ is a sufficient statistic for intent interpretation, i.e., whether:

$$P(\hat{\imath}_t \mid \Omega_t) = P(\hat{\imath}_t \mid S(x_t)) \tag{6}$$

holds in general. If not, characterize the conditions under which this equality fails and quantify the resulting information loss.

4 Theoretical Distinction: Why $S(x)$ Does Not Imply $I(x, C, G, H)$

4.1 The Fundamental Insufficiency Proposition

Proposition 1 (Insufficiency of Sentiment for Intent)**.** *In general:*

$$S(x) \not\Rightarrow I(x, C, G, H) \tag{7}$$

That is, knowledge of $S(x)$ does not uniquely determine $I(x, C, G, H)$. Formally:

$$\exists\, x \in \mathcal{X}, \quad \exists\, (C, G, H) \neq (C', G', H') \;\; \textit{such that} \;\; I(x, C, G, H) \neq I(x, C', G', H') \tag{8}$$

while $S(x)$ remains identical across both contextual configurations.

Proof Sketch. Fix $x =$ "I think I understand now." Any standard sentiment classifier assigns x a positive or neutral label, since the proposition expresses comprehension with no negative lexical signal. Now consider two configurations:

Configuration 1. C_1: the system explained a concept twice; the user answered incorrectly on both prior attempts. G_1: master the concept before an interview. H_1: a history of two failed attempts. Here $I(x, C_1, G_1, H_1)$ is a *face-saving assertion*—the user indicates comprehension to avoid further remediation, while behavioral history reveals incomplete understanding. The pragmatic intent is: `act_type = face-save; underlying_need = further_instruction`.

Configuration 2. C_2: the system explained once; the user answered correctly immediately. G_2: verify comprehension. H_2: one successful answer. Here $I(x, C_2, G_2, H_2)$ is a *genuine confirmation*: `act_type = confirm; underlying_need = proceed`.

Since $S(x)$ is identical in both configurations yet $I(x, C_1, G_1, H_1) \neq I(x, C_2, G_2, H_2)$, it follows that $S(x)$ does not determine $I(x, C, G, H)$. □

Corollary 1.1. *Any architecture that exposes only $S(x)$ as the user-understanding signal loses the information carried in (C, G, H), yielding intent ambiguity irresolvable without access to those variables.*

4.2 Information-Theoretic Perspective

Define the mutual information between the predicted intent $\hat{\imath}$ and an available signal ψ as:

$$\mathrm{MI}(\hat{\imath};\, \psi) = \sum_{i,\,\psi} P(\hat{\imath}{=}i,\; \psi{=}\psi) \cdot \log \frac{P(\hat{\imath}{=}i,\; \psi{=}\psi)}{P(\hat{\imath}{=}i) \cdot P(\psi{=}\psi)} \tag{9}$$

Claim 1. *For systems where intent categories are not uniformly predictable from sentiment alone:*

$$\mathrm{MI}(\hat{\imath};\; S(x)) \;<\; \mathrm{MI}(\hat{\imath};\; I(x, C, G, H)) \tag{10}$$

This formalizes the information gap: sentiment-only systems operate on a compressed, lower-information representation of user state. The equality in Equation (6) fails whenever C, G, or H carry intent-relevant information not captured by $S(x)$.

5 Failure Modes of Sentiment-Based Systems

We identify four canonical failure modes that arise when intent interpretation is reduced to sentiment detection.

5.1 Failure Mode 1: Affective Masking

Definition 5.1 (Affective Masking)**.** Affective masking occurs when a user's surface sentiment misrepresents their underlying intentional state:

$$S(x) = a^* \quad \wedge \quad I(x, C, G, H) \text{ encodes affect } a \neq a^* \tag{11}$$

This is documented in social psychology as *emotional display rules* [11] and *impression management* [15]. In evaluative Human–AI settings, users frequently present positively while experiencing frustration or anxiety because negative self-presentation carries perceived social cost. A sentiment-only system receives uniformly positive signal and responds with affirmation, while the user's true need is diagnostic feedback.

5.2 Failure Mode 2: Sarcasm and Pragmatic Inversion

Definition 5.2 (Pragmatic Inversion)**.** Pragmatic inversion occurs when the literal sentiment of an utterance is inverted by pragmatic context:

$$S(x) = \text{positive} \quad \wedge \quad I(x, C, G, H) \text{ encodes negative intent (or vice versa)} \tag{12}$$

The canonical example is sarcasm: "Oh great, another question I have never seen before" carries positive lexical markers but encodes negative affect and a directive intent. While sarcasm detection has been studied as a standalone task [14, 29], it is rarely integrated as a correction layer in deployed sentiment-driven systems, leaving pragmatic inversions systematically misclassified.

5.3 Failure Mode 3: Temporal Goal Drift

Definition 5.3 (Temporal Goal Drift)**.** User goals evolve across the interaction horizon. Letting G_t denote goal state at time t:

$$G_t \neq G_{t-k} \quad \text{for sufficiently large } k \tag{13}$$

A sentiment classifier $S(x_t)$ provides no information about goal trajectory. A system relying solely on $S(x_t)$ cannot distinguish between a user whose goals have shifted (requiring a strategy change) and one who maintains stable goals but exhibits variable momentary affect. This failure mode is especially consequential in longitudinal platforms where session-level goal evolution is the primary signal of user progress.

5.4 Failure Mode 4: Communicative Role Ambiguity

Definition 5.4 (Role Ambiguity)**.** A single utterance may simultaneously fulfill multiple speech act roles. Sentiment detection collapses this multi-role structure to a single affective label,

discarding illocutionary force. For an utterance exhibiting role ambiguity:

$$I(x, C, G, H) = \{ \text{directive: request_clarification}, \\ \text{expressive: frustration}, \\ \text{commissive: intent_to_retry} \} \tag{14}$$

$$S(x) = \text{negative} \quad [\text{captures expressive only}]$$

A system responding to negative sentiment with emotional support misses the directive and commissive dimensions, producing an affectively appropriate but communicatively inadequate response.

6 Case Study: Sentiment-First Architecture in Longitudinal Systems

6.1 MindNook: Sentiment-Aware Reflective Diary Application

System Description. MindNook is a sentiment-aware digital diary designed to help users track emotional states and reflect on their experiences over time. The system performs linguistic sentiment analysis on diary entries and maintains a temporal record of mood trajectories. Based on detected sentiment patterns, it provides supportive content such as short uplifting stories, reflective prompts, or motivational quotes intended to stabilize or improve mood. The system's architecture conditions its responses primarily on detected sentiment polarity and recent emotional trends derived from user text.

Observed Challenges.

Ambiguity in Sentiment Representation. Diary entries often contain mixed or layered emotions—for example, sadness combined with acceptance, relief, or growth. Mapping such entries to a single polarity score collapses emotional complexity and may reduce the appropriateness of generated support.

Misinterpretation of Positive Reframing. Users describing stressful experiences with positive framing—a known cognitive reappraisal strategy in emotion regulation research [17]—were sometimes interpreted as experiencing stable well-being. In such cases, the system reduced supportive interventions despite continued emotional processing needs.

Contextually Misaligned Support Content. Entries such as "I finally finished everything I needed to do today" were frequently classified as positive and followed by celebratory content. However, in longitudinal context, similar entries often reflected exhaustion rather than achievement satisfaction. Because the system conditions responses primarily on surface sentiment, it may misalign support with the user's implicit intent.

System Design Insight. MindNook illustrates a core structural limitation of sentiment-first architectures in longitudinal Human–AI systems. Emotional polarity alone does not reliably encode user goals, coping strategies, or implicit communicative intent. Without modeling contextual history and evolving user objectives, sentiment-aware systems risk systematic misalignment

between user expression and system response. This directly motivates the formalization of $I(x, C, G, H)$ in Section 3 and the MLIM framework proposed in Section 7.

7 Proposed Multi-Layer Intent Modeling (MLIM) Framework

7.1 Framework Overview

We propose the Multi-Layer Intent Modeling (MLIM) framework as a principled architecture for intent-aware Human–AI communication. MLIM comprises four processing layers that progressively enrich user-input representation from surface affect to full pragmatic intent. Figure 1 illustrates the overall architecture.

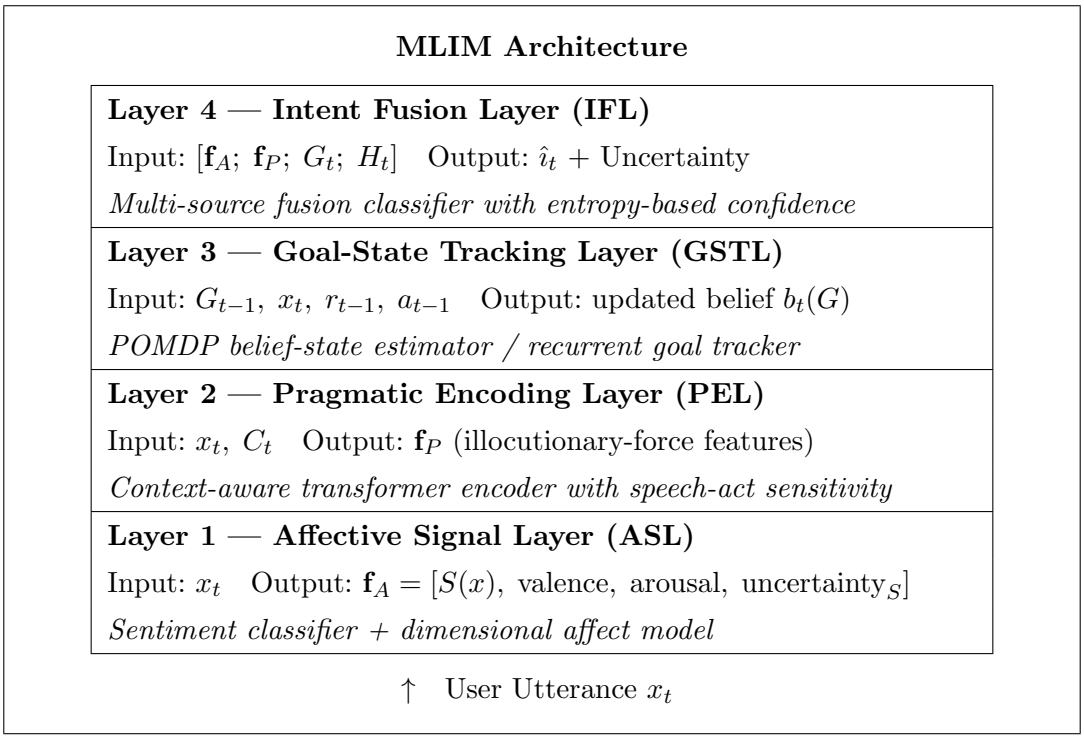

Figure 1: The Multi-Layer Intent Modeling (MLIM) framework. Data flows upward through four processing layers: Layer 1 extracts affective features, Layer 2 encodes pragmatic and illocutionary context, Layer 3 maintains a dynamic goal-state belief, and Layer 4 fuses all signals into a final intent prediction with uncertainty quantification. Replace this schematic with a vector diagram before arXiv submission.

The four layers are:

Layer 1 — Affective Signal Layer (ASL). Processes raw utterance x to produce affective features $\mathbf{f}_A$:

$$\mathbf{f}_A = \varphi_A(x) = \big[S(x),\ \text{valence}(x),\ \text{arousal}(x),\ \text{uncertainty}_S(x)\big] \tag{15}$$

This layer treats $S(x)$ as one signal among several, not as the sole user representation.

Layer 2 — Pragmatic Encoding Layer (PEL). Applies a speech-act-aware contextual

encoder:

$$\mathbf{f}_P = \varphi_P(x, C) = \text{Encoder}_P([x;\ C]) \tag{16}$$

where Encoder_P produces representations sensitive to illocutionary force (directive, expressive, commissive, representative) in context C.

Layer 3 — Goal-State Tracking Layer (GSTL). Maintains a dynamic estimate of user goal state:

$$G_t = \text{GoalTracker}(G_{t-1},\ x_t,\ r_{t-1},\ a_{t-1}) \tag{17}$$

This layer may be implemented as a belief-state estimator in a POMDP framework [50] where G is a latent variable inferred from observable interaction signals.

Layer 4 — Intent Fusion Layer (IFL). Integrates outputs from Layers 1–3 with interaction history:

$$\hat{\imath}_t = \text{IntentClassifier}([\mathbf{f}_A;\ \mathbf{f}_P;\ G_t;\ H_t]) \tag{18}$$

7.2 Integration with Existing Dialogue Architectures

MLIM is modular and compatible with existing architectures. In standard pipeline dialogue systems [50], MLIM can augment or partially replace the NLU module, with Layer 1 feeding affective response generation while Layers 2–4 provide enriched signals to the dialogue state tracker and policy module. In end-to-end neural dialogue systems [46], MLIM layers function as additional conditioning signals in the encoder-decoder architecture.

8 Mathematical Formulation of Intent-Aware Modeling

8.1 Joint Intent-Affect Posterior

The full intent-aware model targets:

$$P(\hat{\imath}_t,\ \hat{a}_t \mid x_t, C_t, G_t, H_t) \tag{19}$$

By the chain rule of probability:

$$P(\hat{\imath}_t, \hat{a}_t \mid x_t, C_t, G_t, H_t) = P(\hat{\imath}_t \mid \hat{a}_t, x_t, C_t, G_t, H_t) \cdot P(\hat{a}_t \mid x_t, C_t, H_t) \tag{20}$$

This factorization makes explicit that intent is conditioned on affect—not vice versa—and that affect is conditioned on context and history, not on utterance alone.

8.2 Multi-Source Feature Fusion

Let $\mathbf{f} = [\mathbf{f}_A;\ \mathbf{f}_P;\ \mathbf{f}_G;\ \mathbf{f}_H]$ denote the concatenated feature vector from all four MLIM layers. The intent classifier implements:

$$P(\hat{\imath}_t = i \mid \mathbf{f}) = \text{softmax}(W_I \cdot \mathbf{f} + \mathbf{b}_I)_i \tag{21}$$

where $W_I \in \mathbb{R}^{|\mathcal{I}| \times d_f}$, $\mathbf{b}_I \in \mathbb{R}^{|\mathcal{I}|}$, and d_f is the total feature dimensionality. The joint training objective is:

$$\mathcal{L}(\theta) = -\sum_t \Big[\lambda_1 \cdot \log P(\hat{\imath}_t \mid \mathbf{f}_t; \theta) + \lambda_2 \cdot \log P(\hat{a}_t \mid x_t, C_t; \theta)\Big] + \Omega(\theta) \tag{22}$$

where λ_1, λ_2 are loss-weighting hyperparameters and $\Omega(\theta)$ is a regularization term.

8.3 Goal-State Belief Estimation

The GSTL module implements a belief update over $\mathcal{G}$:

$$b_t(g) = P(G_t{=}g \mid x_{1:t}, r_{1:t-1}, a_{1:t-1}) \;\propto\; P(x_t \mid G_t{=}g, C_t) \sum_{g'} P\big(G_t{=}g \mid G_{t-1}{=}g'\big)\, b_{t-1}(g') \tag{23}$$

This is a Hidden Markov Model formulation with transition model $P(G_t \mid G_{t-1})$ capturing goal drift and observation model $P(x_t \mid G_t, C_t)$ capturing how goals shape utterance generation [50].

8.4 Intent Confidence and Uncertainty Quantification

For each predicted intent $\hat{\imath}_t$, the system computes:

$$\mathrm{Conf}(\hat{\imath}_t) = \max_i \; P(\hat{\imath}_t = i \mid \mathbf{f}_t) \tag{24}$$

$$\mathrm{Uncertainty}(\hat{\imath}_t) = -\sum_i P(\hat{\imath}_t = i \mid \mathbf{f}_t) \log P(\hat{\imath}_t = i \mid \mathbf{f}_t) \tag{25}$$

High entropy signals the system to solicit clarification rather than act on a low-confidence prediction, preventing commitment to an incorrect intent interpretation with potentially harmful downstream consequences.

8.5 Reduction to Sentiment-Only as a Special Case

The sentiment-only system is a strict special case of MLIM in which Layers 2–4 are disabled:

$$\mathbf{f} = \mathbf{f}_A = S(x), \qquad C_t = \emptyset, \qquad G_t = \emptyset, \qquad H_t = \emptyset \tag{26}$$

Under this reduction, $P(\hat{\imath}_t \mid \mathbf{f}) = P(\hat{\imath}_t \mid S(x_t))$, which Proposition 1 establishes to be strictly less informative than the full posterior $P(\hat{\imath}_t \mid x_t, C_t, G_t, H_t)$ whenever intent varies independently of sentiment.

9 Comparison: Sentiment Detection vs. Intent Interpretation

Table 1 provides a structured comparison across key theoretical, computational, and application dimensions.

Table 1: Structured comparison of Sentiment Detection $S(x)$ and Intent Interpretation $I(x, C, G, H)$ across theoretical, computational, and application dimensions.

Dimension	**Sentiment Detection** $S(x)$	**Intent Interpretation** $I(x, C, G, H)$
Primary Signal	Emotional polarity (pos/neg/neutral)	Goal-directed meaning in pragmatic context
Input Space	Surface text tokens x	x + context C + goal G + history H
Output Space	Affective label or scalar score	Structured intent class with confidence
Temporal Scope	Momentary (single utterance)	Longitudinal (session- and user-level)
Context Sensitivity	Low (lexical-semantic only)	High (pragmatic, situational, relational)
Ambiguity Handling	Via polarity score only	Disambiguation via contextual grounding
Sarcasm / Irony	High failure rate	Partially recoverable via discourse context
Affective Masking	High failure rate (undetectable)	Recoverable via history-conditioned inference
Goal Tracking	Not supported	Explicit via GSTL module
Speech Act Modeling	Not supported	Explicit via Pragmatic Encoding Layer
Application Scope	Content moderation, reviews	Coaching, dialogue management, mental health
Computational Cost	Low (single classifier pass)	Higher (multi-signal fusion)
Formal Sufficiency	Necessary but insufficient	Sufficient only with full contextual modeling

10 Research Agenda and Open Problems

We identify six open research problems whose resolution is necessary for advancing intent-aware Human–AI communication.

Open Problem 1: Context Horizon Selection. The context window $C_t = (x_{t-k}, \ldots, x_{t-1})$ is parameterized by horizon k. Small k loses relevant pragmatic context; large k introduces noise and computational cost. Principled methods for adaptive k selection—informed by dialogue structure detection [18] or attention-based relevance scoring—remain open.

Open Problem 2: Goal Inference from Implicit Signals. When users do not state their goals explicitly, inference must rely on implicit behavioral signals—response latency, revisitation patterns, query reformulation. Learning reliable goal inference models without explicit annotation is a fundamental open challenge.

Open Problem 3: Multi-Modal Intent Grounding. In voice and embodied interfaces, intent is distributed across linguistic, prosodic, and gestural channels. Extending MLIM to incorporate acoustic features (speech rate, pitch contour, pause structure) with textual representations remains an active challenge [3, 34].

Open Problem 4: Out-of-Distribution Intent Detection. In open-domain systems, users will express intents outside the training distribution. Robust out-of-distribution detection analogous to out-of-scope detection [23] must be extended to the multi-signal, contextual setting.

Open Problem 5: Evaluation Metrics for Intent Interpretation. Standard NLU benchmarks evaluate single-utterance, closed-label classification. Evaluating intent interpretation across sessions, under evolving goals and affective masking, requires new protocols and datasets that do not yet exist at scale.

Open Problem 6: Low-Resource and Cross-Cultural Intent Modeling. Intent expression is culturally conditioned [19, 27]. Cross-cultural intent modeling requires multilingual, multicultural corpora and culturally-aware frameworks that remain substantially underdeveloped.

11 Ethical and Privacy Considerations

11.1 Privacy and User Data

MLIM requires maintaining interaction history H and goal state G across sessions, introducing substantially greater privacy risk than utterance-level sentiment analysis. Systems must apply data minimization principles [12], enforce purpose limitation, and provide users with transparent access to and deletion of stored data. Differential privacy [10] should be considered for goal-state inference models. Federated learning [28] offers a principled direction for personalizing intent models without centralizing sensitive longitudinal data.

11.2 Intent Model Transparency and Explainability

When a system acts on inferred intent $\hat{\imath}_t$, users have a reasonable expectation of understanding why. Attention-based explanations [47], gradient-based attributions [36], and counterfactual explanations [44] are candidate approaches, though their sufficiency for intent explanation remains contested.

11.3 Risk of Intent Misattribution

Incorrect intent inference in mental health or coaching contexts may cause active harm. A system that falsely infers suicidal ideation from an ambiguous entry violates privacy; one that misses genuine distress due to positive framing may withhold needed support. Systems must implement uncertainty thresholds (Section 8.4) and human-in-the-loop escalation protocols for high-stakes intent categories.

11.4 Bias and Demographic Fairness

Sentiment classifiers exhibit demographic bias across groups, dialects, and writing styles [22]. Intent models trained on similar corpora are likely to inherit and amplify these biases, as Layer 1 of MLIM uses affective features as input. Fairness auditing across demographic groups is a necessary prerequisite for any MLIM-style deployment.

12 Conclusion

This paper has established, on formal theoretical and system-observational grounds, that sentiment detection is a necessary but strictly insufficient mechanism for intent interpretation in Human–AI communication. We demonstrated this through formal definitions of $S(x)$ and $I(x, C, G, H)$, a proof that $S(x)$ does not imply $I(x, C, G, H)$ in the general case, and the identification of four structural failure modes: affective masking, pragmatic inversion, temporal goal drift, and communicative role ambiguity.

System-level observations from MindNook grounded these distinctions in real interaction patterns, demonstrating that sentiment-first architectures produce systematic response mismatches in longitudinal, reflective Human–AI dialogue contexts—particularly where users engage in positive reframing, mixed-affect expression, or implicit goal signaling. The Multi-Layer Intent Modeling (MLIM) framework integrates affective, pragmatic, goal-theoretic, and historical signals into a unified intent-aware architecture.

The open research agenda—spanning context horizon selection, implicit goal inference, multimodal grounding, out-of-distribution detection, evaluation methodology, and cross-cultural modeling—outlines substantial work required before intent-aware Human–AI communication achieves the reliability necessary for high-stakes deployment. The ethical dimensions of longitudinal user modeling, intent transparency, and demographic fairness must be addressed in parallel with technical development.

We anticipate that the formalization developed here will provide a useful conceptual foundation for subsequent empirical work in intent-aware dialogue systems, and that the MLIM framework will serve as a tractable architecture for such investigations.

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